Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Agentic AI Automation with n8n

TGS Ref No: TGS-2023035977

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 3.2

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	24 June 2026	First version — 3-day lesson plan aligned to the agentic n8n course flow	Tertiary Infotech Academy Pte Ltd
2.0	26 June 2026	Renumbered Day 2 activities: Investment Advisor → Activity 5, Finance Advisor → Activity 6, RAG → Activity 7; updated Activity 8 to the integrated HR Service Portal (Leave Approval, Dashboard Data, AI Chatbot with Guardrails)	Tertiary Infotech Academy Pte Ltd
3.0	1 July 2026	Replaced Activity 7 with two new RAG labs — Activity 7a (RAG chatbot: web PDF upload → vector store → Telegram Q&A) and Activity 7b (customer-support RAG agent across three vector databases: Supabase, Pinecone, Qdrant)	Tertiary Infotech Academy Pte Ltd
3.1	1 July 2026	Added a page-numbered Table of Contents; verified daily-schedule table borders and footer page numbers; aligned the Lesson Plan with the slides, Learner Guide and lab workflows (activities 1–8 + capstone)	Tertiary Infotech Academy Pte Ltd
3.2	17 Oct 2026	Aligned with slide deck v48: learners sign in with assigned n8n training accounts (no self sign-up) and use the free n8n Gateway credits for AI agents (API keys optional); Topic 1 now opens with why agentic AI and workflow first vs agent first before n8n is introduced; Activity 4a is the HR Policy Chatbot built with n8n Assistant; the mini	Tertiary Infotech Academy Pte Ltd

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		capstone is built with n8n Assistant	

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Course Title	Agentic AI Automation with n8n
Course Code	TGS-2023035977
Duration	3 Days (24 training hours)
Daily Schedule	9:30 AM – 6:30 PM (8 training hours/day, excluding lunch)
Lunch Break	1:00 PM – 2:00 PM (1 hour)
Breaks	Short tea breaks are scheduled within each day's training hours
Delivery Mode	Instructor-led, hands-on labs with group work and presentations
Prerequisites	Basic computer literacy; a Google/Gmail account; no coding experience required
Tools & Accounts	n8n (assigned training accounts + local Docker), free n8n Gateway credits (API keys optional), Telegram, Gmail/Outlook, Google Sheets, Twelve Data & NewsAPI, Claude Code

Course Overview

This 3-day hands-on course teaches participants to design, build and secure agentic AI automations with n8n. Learners set up n8n (trial and local Docker), automate forms and email, store data in Data Tables and external spreadsheets, build Telegram-triggered AI agents, integrate webhooks and external APIs, add Retrieval-Augmented Generation (RAG), and apply human-in-the-loop guardrails — culminating in a mini capstone project that is presented and assessed.

Learning Outcomes

By the end of this course, participants will be able to:

- Set up n8n using a cloud trial and a local Docker Compose installation, and navigate the n8n UI.
- Apply the core concepts of triggers, actions, nodes and flows to build automated workflows.
- Automate form submissions with QR codes, send admin emails and capture data in Data Tables and Google Sheets/Excel.
- Design Telegram-triggered AI agents using LLMs, memory, tools and system instructions.
- Expose workflows via webhooks and build website chatbots that respond to live browser requests.
- Integrate external services through APIs and HTTP requests, including real-time market and news data.
- Implement Retrieval-Augmented Generation (RAG) and route an agent between multiple data sources.
- Apply human-in-the-loop and guardrail patterns to make AI automations safe and reliable.
- Plan, build, present and self-assess an end-to-end mini capstone automation project.

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Day 1

Workflow Automation with n8n · AI Agents

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration
9:30 – 9:45	Welcome, course objectives & overview (TGS-2023035977)	15 min
9:45 – 1:00	**Topic 1: Workflow Automation with n8n**	—
9:45 – 10:15	Why agentic AI, and why workflow first — history of AI, generative vs agentic AI, agentic AI vs AI agents, workflow first vs agent first, business value, AI security; overview of n8n	30 min
10:15 – 10:35	Set up n8n — sign in with the assigned n8n training accounts, claim the free AI (Gateway) credits, n8n Assistant demo; local install with Docker Compose shown	20 min
10:35 – 10:50	Explore the n8n UI	15 min
10:50 – 11:10	Key concepts of n8n — Triggers, Actions, Nodes, Flows	20 min
11:10 – 11:35	Activity 1: QR Code form (name, email, phone, message) + email to admin via Gmail/Outlook (2 nodes)	25 min
11:35 – 12:25	Group Activity (3–4 pax): design an event flyer (e.g. bowling), generate a QR code with the QR generator, build the ad, then present (review past-student examples)	50 min
12:25 – 1:00	Activity 2: Add an n8n Data Table to Activity 1 to capture submissions (3 nodes)	35 min
Time	**Topic / Activity**	**Duration**
1:00 – 2:00	***Lunch Break***	60 min

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration
2:00 – 2:40	Activity 3a: Conditional response — if yes, save the date to the Data Table; if no, send a thank-you email (note: trial Data Table data is not persistent)	40 min
2:40 – 3:20	Activity 3b: Conditional response — replace the Data Table with Google Sheet/Excel for persistent storage (Google training accounts); if no, send a thank-you email	40 min

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3:20 – 3:35	*Tea Break*	15 min
3:35 – 6:05	**Topic 2: AI Agents**	—
3:35 – 4:05	Overview of AI Agents — LLM, Memory, Tools and System Instruction	30 min
4:05 – 4:50	Activity 4a: Telegram HR Policy Chatbot (Ava) — create the bot with BotFather, build it with n8n Assistant from one prompt, connect, publish and chat (free n8n Gateway credits)	45 min
4:50 – 6:05	Activity 4b: Give Ava a tool — attach an Employees Data Table (reuse Activity 2 data or upload a 100-record employee CSV) so she answers from real staff records	75 min
6:05 – 6:30	Day 1 recap, Q&A and wrap-up	25 min

Day 2

Webhooks · APIs & HTTP Requests · RAG

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration
9:30 – 9:45	Day 1 recap & Day 2 objectives	15 min
9:45 – 12:10	**Topic 3: Webhooks**	—
9:45 – 10:15	Overview of Webhooks — use cases and external triggers	30 min
10:15 – 11:35	Activity 5: Website chatbot via webhook — follow the n8n Investment Advisor reference site (CORS, Webhook trigger, Respond to Webhook, AI Agent)	80 min
11:35 – 11:50	*Tea Break*	15 min
11:50 – 12:10	Selected learners present their live website and chatbot	20 min
12:10 – 1:00	**Topic 4: APIs and HTTP Requests**	—
12:10 – 12:40	Overview of APIs and HTTP Requests — methods, headers, query parameters, credentials	30 min
12:40 – 1:00	Activity 6 setup: sign up for Twelve Data and NewsAPI; copy API keys; import the workflow	20 min
Time	**Topic / Activity**	**Duration**
1:00 – 2:00	*Lunch Break*	60 min

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration
2:00 – 4:00	**Topic 4 (cont.): APIs and HTTP Requests**	—
2:00 – 3:20	Activity 6 (cont.): Configure	80 min

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	Twelve Data and NewsAPI keys; build the Finance API → Telegram day-trading agent (candles1min / candles15min / candles1hr + news, AI Agent, Telegram reply)	
3:20 – 4:00	Overview of Retrieval-Augmented Generation (RAG) — Tokenization, Embeddings, Vector Stores	40 min
4:00 – 4:15	***Tea Break***	15 min
4:15 – 6:30	**Topic 5: Retrieval-Augmented Generation (RAG)**	—
4:15 – 4:45	RAG ingestion walkthrough — the upload → embed → vector-store → retrieve loop; set up OpenAI / Gemini embedding credentials	30 min
4:45 – 6:05	Activity 7a: RAG chatbot — upload an IT-Support FAQ PDF from a web page into a vector store, then ask the Telegram bot (Gemini embeddings + knowledge_base tool). Activity 7b: customer-support RAG agent for a training center, ingesting 20 course brochures across three vector databases (Supabase pgvector, Pinecone, Qdrant) answering via a website chat widget	80 min
6:05 – 6:30	Day 2 recap, Q&A and wrap-up	25 min

Day 3

Security & Guardrails · Mini Capstone Project

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration
9:30 – 9:45	Day 2 recap & Day 3 objectives	15 min
9:45 – 1:00	**Topic 6: Security and Guardrails — HR Service Portal**	—
9:45 – 10:25	Overview of Human-in-the-Loop and the HR Service Portal — three coordinated n8n workflows behind one web page (Leave Application, Dashboard, AI Chatbot)	40 min
10:25 – 11:25	Activity 8a: Leave Application with Human-in-the-Loop — webhook receives leave request, emails manager an Approve/Reject button (Send and Wait for Response), emails employee the outcome	60 min

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11:25 – 11:40	***Tea Break***	15 min
11:40 – 12:00	Selected learners present their Activity 8a approval flow	20 min
12:00 – 12:25	Overview of Guardrails — pre/post patterns, blocked responses, canned-reply fallbacks	25 min
12:25 – 1:00	Activity 8b & 8c: Build the HR Dashboard Data webhook (leave-balance stats) and the AI Chatbot with Input & Output Guardrails; wire all three endpoints in the HR Service Portal	35 min
Time	**Topic / Activity**	**Duration**
1:00 – 2:00	***Lunch Break***	60 min

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration
2:00 – 6:30	**Topic 7: Mini Capstone Project**	—
2:00 – 2:30	Capstone briefing — requirements, scope and assessment criteria; AI-assisted workflow creation with n8n Assistant (example prompt: end-to-end video creation — topic, research, script, production, publish)	30 min
2:30 – 4:15	Mini capstone build (supervised lab time) — describe the process to n8n Assistant, review and connect credentials, test and add a guardrail or approval step	105 min
4:15 – 4:30	***Tea Break***	15 min
4:30 – 5:45	Presentation of mini capstone projects	75 min
5:45 – 6:30	Assessment and course closing	45 min

Tools, Accounts & Resources

QR Code Generator	https://alfredang.github.io/qrcodegenerator/
Webhook reference — Activity 5: Investment Advisor	https://alfredang.github.io/n8n-investmentadvisor/
Finance API reference — Activity 6: Finance Advisor	https://alfredang.github.io/n8n-financeadvisor/
Finance Data API — Twelve Data	https://twelvedata.com/login
News API — NewsAPI	https://newsapi.org/
LLM access	Free n8n Gateway credits on each training account; OpenAI / Google Gemini API keys optional
Local n8n	Docker Compose self-hosted install (see Learner Guide)

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Assessment

Participants are assessed through their hands-on activities across the three days and a mini capstone project on Day 3. The capstone is presented to the class and evaluated against the course learning outcomes — workflow design, AI agent integration, webhook and API use, Retrieval-Augmented Generation, and the application of security guardrails.

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